Final Project Datasets

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Dataset 1: Recipes and Ratings

This dataset contains recipes and ratings from food.com. It was originally scraped and used by the authors of this recommender systems paper.

Getting the Data

Access the Recipes and Ratings dataset via Google Drive. You'll download two CSV files:

- `RAW_recipes.csv` contains recipes.
- `RAW_interactions.csv` contains reviews and ratings submitted for the recipes in `RAW_recipes.csv`.

We've provided you with a subset of the raw data used in the original report, containing only the recipes and reviews posted since 2008, since the original data is quite large.

A description of each column in both datasets is given below.

Recipes

For context, you may want to look at an example recipe directly on food.com.

Column	Description
`'name'`	Recipe name
`'id'`	Recipe ID
`'minutes'`	Minutes to prepare recipe
`'contributor_id'`	User ID who submitted this recipe
`'submitted'`	Date recipe was submitted
`'tags'`	Food.com tags for recipe
`'nutrition'`	Nutrition information in the form [calories (#), total fat (PDV), sugar (PDV), sodium (PDV), protein (PDV), saturated fat (PDV), carbohydrates (PDV)]; PDV stands for "percentage of daily value"
`'n_steps'`	Number of steps in recipe
`'steps'`	Text for recipe steps, in order
`'description'`	User-provided description

Ratings

Column	Description
`'user_id'`	User ID
`'recipe_id'`	Recipe ID
`'date'`	Date of interaction
`'rating'`	Rating given
`'review'`	Review text

After downloading the datasets, you **must** follow the following steps to merge the two datasets and create a column containing the average rating per recipe:

1. Left merge the recipes and interactions datasets together.
2. In the merged dataset, fill all ratings of 0 with `np.nan`. (Think about *why* this is a reasonable step, and include your justification in your website.)
3. Find the average rating per recipe, as a Series.
4. Add this Series containing the average rating per recipe back to the recipes dataset however you'd like (e.g., by merging). **Use the resulting dataset for all of your analysis.** (For the purposes of Project 4, the `'review'` column in the interactions dataset doesn't have much use.)

Example Questions and Prediction Problems

Feel free to base your exploration into the dataset in Steps 1-4 around one of these questions, or come up with a question of your own.

- What types of recipes tend to have the most calories?
- What types of recipes tend to have higher average ratings?
- What types of recipes tend to be healthier (i.e. more protein, fewer carbs)?
- What is the relationship between the cooking time and average rating of recipes?

Feel free to use one of the prompts below to build your predictive model in Steps 5-8, or come up with a prediction task of your own.

- Predict ratings of recipes.
- Predict the number of minutes to prepare recipes.
- Predict the number of steps in recipes.
- Predict calories of recipes.

Make sure to justify what information you would know at the "time of prediction" and to only train your model using those features.

Special Considerations

Step 2: Data Cleaning and Exploratory Data Analysis

Some columns, like `'nutrition'`, contain values that look like lists, but are actually strings that look like lists. You may want to turn the strings into actual lists, or create columns for every unique value in those lists. For instance, per the data dictionary, each value in the `'nutrition'` column contains information in the form `"[calories (#), total fat (PDV), sugar (PDV), sodium (PDV), protein (PDV), saturated fat (PDV), and carbohydrates (PDV)]"`; you could create individual columns in your dataset titled `'calories'`, `'total fat'`, etc.

Step 4: Assessment of Missingness

There are only three columns in the merged dataset that contain missing values; make sure you're using the merged dataset for all of your analysis (and that you followed the steps at the top of this page exactly).

Dataset 2: League of Legends

Welcome to Summoner's Rift! This dataset contains information of players and teams from over 10,000 League of Legends competitive matches.

You'll probably want to be at least a little bit familiar with *League of Legends* and its terminology to use this dataset. If not, one of the other datasets may be more interesting to you.

Getting the Data

The data can be found on the website Oracle's Elixir at the provided Google Drive link.

We've verified that it's possible to satisfy the requirements of the project using match data from 2022. You're welcome to use newer or older datasets if you wish, but keep in mind that League of Legends changes significantly between years; this can make it difficult to combine or make comparisons between datasets from different years.

Example Questions and Prediction Problems

Feel free to base your exploration into the dataset in Steps 1-4 around one of these questions, or come up with a question of your own.

- Looking at tier-one professional leagues, which league has the most "action-packed" games? Is the amount of "action" in this league significantly different than in other leagues? Note that you'll have to come up with a way of quantifying "action".
- Which competitive region has the highest win rate against teams outside their region? Note you will have to find and merge region data for this question as the dataset does not have it.
- Which role "carries" (does the best) in their team more often: ADCs (Bot lanes) or Mid laners?

- Is Talon (former tutor Costin's favorite champion) more likely to win or lose any given match?

Feel free to use one of the prompts below to build your predictive model in Steps 5-8, or come up with a prediction task of your own.

- Predict if a team will win or lose a game.
- Predict which role (top-lane, jungle, support, etc.) a player played given their post-game data.
- Predict how long a game will take before it happens.
- Predict which team will get the first Baron.

Make sure to justify what information you would know at the "time of prediction" and to only train your model using those features.

Special Considerations

Step 2: Data Cleaning and Exploratory Data Analysis

- Each `'gameid'` corresponds to up to 12 rows – one for each of the 5 players on both teams and 2 containing summary data for the two teams (try to find out what distinguishes those rows). After selecting your line of inquiry, make sure to remove either the player rows or the team rows so as not to have issues later in your analysis.
- Many columns should be of type `bool` but are not.

Dataset 3: Power Outages

This dataset has major power outage data in the continental U.S. from January 2000 to July 2016.

Getting the Data

The dataset is available via Purdue University.

Note: If you are having a hard time with the "This dataset" link, hold shift and click the link to open it into a new tab and then refresh that new tab.

A data dictionary is available at the article "Data on major power outage events in the continental U.S." under *Table 1. Variable descriptions*.

Example Questions and Prediction Problems

Feel free to base your exploration into the dataset in Steps 1-4 around one of these questions, or come up with a question of your own.

- Where and when do major power outages tend to occur?
- What are the characteristics of major power outages with higher severity? Variables to consider include location, time, climate, land-use characteristics, electricity consumption patterns, economic characteristics, etc. What risk factors may an energy company want to look into when predicting the location and severity of its next major power outage?
- What characteristics are associated with each category of cause?
- How have characteristics of major power outages changed over time? Is there a clear trend?

Feel free to use one of the prompts below to build your predictive model in Steps 5-8, or come up with a prediction task of your own.

- Predict the severity (in terms of number of customers, duration, or demand loss) of a major power outage.
- Predict the cause of a major power outage.
- Predict the number and/or severity of major power outages in the year 2022.
- Predict the electricity consumption of an area.

Make sure to justify what information you would know at the "time of prediction" and to only train your model using those features.

Special Considerations

Step 2: Data Cleaning and Exploratory Data Analysis

- The data is given as an Excel file rather than a CSV. Open the data in Google Sheets or another spreadsheet application and determine which rows and columns of the sheet should be ignored when loading the data in `pandas`.
    - Note that `pandas` can load multiple filetypes: `pd.read_csv`, `pd.read_excel`, `pd.read_html`, `pd.read_json`, etc.
- The power outage start date and time is given by `'OUTAGE.START.DATE'` and `'OUTAGE.START.TIME'`. It would be preferable if these two columns were combined into one `pd.Timestamp` column. Combine `'OUTAGE.START.DATE'` and `'OUTAGE.START.TIME'` into a new `pd.Timestamp` column called `'OUTAGE.START'`. Similarly, combine `'OUTAGE.RESTORATION.DATE'` and `'OUTAGE.RESTORATION.TIME'` into a new `pd.Timestamp` column called `'OUTAGE.RESTORATION'`.
    - `pd.to_datetime` and `pd.to_timedelta` will be useful here.
- To visualize geospatial data, consider Folium or another geospatial plotting library. You can even embed Folium maps in your website! If `fig` is

a `folium.folium.Map` object, then `fig._repr_html_()` evaluates to a string containing your plot as HTML; use `open` and `write` to write this string to an `.html` file.

Propose Your Own Dataset

Instead of using one of the three pre-approved datasets (Recipes and Ratings, League of Legends, or Power Outages), you may propose your own dataset. However, your dataset **must** be able to support every step of the project. Read the requirements below carefully before proposing.

Dataset Requirements

Your proposed dataset must satisfy **all** of the following:

1. **Publicly available.** The dataset must be freely accessible online (e.g. from Kaggle, a government data portal, an academic paper, etc.). Since your final deliverable includes a public-facing website, anyone should be able to find and download the data you used. Datasets that require special access, private API keys, or paid subscriptions are not allowed.

2. **Sufficiently large.** The dataset must have at least **1,000 rows** (after any necessary cleaning) and at least **8 columns**. A very small dataset will not support meaningful exploratory data analysis, hypothesis testing, or predictive modeling.

3. **Mix of column types.** The dataset must contain both **quantitative** (numerical) and **categorical** (nominal or ordinal) columns. You need this variety to complete the Baseline Model and Final Model steps, which require you to handle different feature types in a `sklearn Pipeline`.

4. **Non-trivial missingness.** At least one column must have a meaningful amount of missing values (not just one or two rows). You need this to complete Step 3: Assessment of Missingness, which requires you to:

    - Reason about whether a column is NMAR.

    - Run permutation tests to determine whether the missingness of a column depends on other columns.

  If your dataset has no missing values at all, it cannot be used for this project.

5. **Supports a hypothesis test.** The dataset must lend itself to at least one interesting hypothesis test or permutation test (beyond missingness). Think about whether you can pose a meaningful question that compares groups or tests an association.

6. **Supports a prediction problem.** The dataset must have a reasonable response variable that you can predict using other columns, either as a **classification** or **regression** task. Make sure there are enough features to build both a baseline model (at least 2 features) and an improved final model (with at least 2 additional engineered features).

7. **Supports a fairness analysis.** The dataset must contain (or allow you to derive) at least two meaningful groups so that you can assess whether your predictive model performs equitably across those groups using a permutation test.

8. **Not a dataset used in DSC 259R lectures, discussions, or other assignments.** Choose something new!

How to Submit Your Proposal

Submit your proposal on Gradescope by **Monday of Week 7 at 11:59PM PST** — this is one week before Checkpoint 1 is due, giving us time to review and give you a decision before the checkpoint deadline. **You may not begin the project with a custom dataset until your proposal is approved.**

Your proposal should include:

1. **Dataset name and link.** Provide the name of the dataset and a direct URL to download it.

2. **Brief description (2-3 sentences).** What does the dataset contain? How was it collected?
3. **Number of rows and columns.** State the size of the dataset.
4. **Column descriptions.** List the columns you plan to use, along with their types (quantitative, nominal, ordinal) and brief descriptions.
5. **Missingness.** Identify which column(s) have non-trivial missingness and briefly explain why you think the missingness exists (i.e. what is the data generating process that leads to these missing values?).
6. **Proposed question.** State the question you plan to investigate in Steps 1-4.
7. **Proposed prediction problem.** State the prediction problem you plan to tackle in Steps 5-8, including the response variable and whether it is a classification or regression task.
8. **Proposed fairness groups.** Describe the two groups you plan to compare in Step 8: Fairness Analysis.

After Approval

Once your proposal is approved, you will follow the exact same project instructions as everyone else — the only difference is the dataset you use. All eight steps (Introduction through Fairness Analysis), the notebook requirements, and the website requirements apply to you in full.

If your proposal is **not** approved, we will let you know why and give you the option to either revise your proposal (if time permits) or choose one of the three pre-approved datasets.

Do not wait until the last minute to submit your proposal. If your proposal is rejected and you need to switch to a pre-approved dataset, you will have less time to complete the project.